UNIT 3

Introduction to digital forensics

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Unit 3	Introduction to Digital Forensics	06 Hours
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What is computer forensic ?

- Computer forensics means checking computers and digital devices to find evidence.
- It includes collecting, saving, studying, and showing computer-related data safely.
- It helps in finding deleted or hidden information from computers, mobiles, hard disks, etc.
- It is also called digital discovery, data recovery, or computer investigation.
- Used in criminal cases, court matters, company disputes, and HR investigations.

Use of computer forensic in law enforcement

- Recover deleted files: Helps police recover deleted documents, photos, and other data.
- Check unallocated space: Finds hidden or leftover data stored in unused parts of a hard disk.
- Trace digital artefacts: Finds small traces left by the operating system and explains what they mean.
- Find hidden files: Detects files that users cannot see but contain information about past activities (like when a file was created, edited, opened, or deleted).
- String searches: Helps search for emails or messages even when there is no proper email app available.

Computer forensic assistance to hr/employment preceedings

How Computer Forensics Helps HR Cases

- Helps in cases like sexual harassment, discrimination, or wrongful termination.
- Evidence can be found in emails, servers, and employee computers.

Employer Safeguard Program

- Employers must protect company data from being deleted or stolen.
- Before firing an employee, a forensic expert should make a full copy of the employee's computer data.
- This protects the company if the employee tries to change or delete files.
- Lost or damaged data can be recovered and restored.

Helps the company defend against false accusations and prove if someone deleted or stole important files.

What Forensics Can Detect

- Websites visited
- Files downloaded
- When files were last opened
- Attempts to hide, destroy, or fake evidence
- Deleted text that still remains in electronic versions
- Backup copies stored in fax machines
- Computer-based faxes stored for long periods
- Emails stored for months or years
- Personal or business financial records saved digitally

Computer forensic Services

1. Data Seizure

- Experts help collect and take control of computers and devices legally.
- They know how to find and secure important digital evidence safely.

2. Data Duplication / Preservation

- They create an exact copy of the original data.
- This protects the original data from any changes.
- Work is done on the copy so the real data stays safe and unchanged.

3. Data Recovery

- Experts recover deleted, hidden, or inaccessible data.
- Their knowledge of storage systems helps them find lost evidence

4. Document Searches

- They can quickly search through thousands of digital documents.
- Makes the investigation faster and less time-consuming.

5. Media Conversion

- Experts take data from old or unreadable devices.
- They convert it into modern formats and move it to new storage for analysis.

6. Expert Witness Services

- Forensic experts can explain technical computer processes in simple language in court.
- Helps judges and juries understand how the evidence was found and why it matters.

7. Computer Evidence Service Options

- Standard Service: Work during regular hours until important data is found.
- On-site Service: Experts come to your location and collect digital evidence directly.
- Emergency Assistance: Immediate, high-priority work in labs until the case needs are met.

Benefits of forensic methodology

- Evidence stays safe: Experts make sure no important data is damaged, deleted, or changed during the investigation.
- No viruses added: They ensure that no computer virus enters the system while analysing it.
- Proper protection of evidence: All extracted data is handled safely and protected from any physical or electronic damage.
- Chain of custody maintained: They keep a clear record of who handled the evidence from start to end, ensuring legal validity.
- Minimal business disruption: The company’s daily work is disturbed as little as possible during the investigation.
- Legal and ethical handling: Any private or confidential information found by mistake is handled legally and not shared with anyone.

Steps taken by computer forensic specialists

1. Protect the computer system:

- Ensure the device is safe from any changes, damage, corruption, or virus infection during investigation.

2. Find all files:

- Locate every type of file—normal files, deleted files, hidden files, password-protected files, and encrypted files.

3. Recover deleted data:

- Bring back files that were deleted but still exist in the system.

4. Check hidden and temporary folders:

- Look inside secret folders, temporary files, and swap files used by apps and the operating system.

5. Open protected files:

- Access password-protected or encrypted files.

6. Examine special storage areas:

- Study data stored in unallocated space and slack space (unused parts of disk where old data may still exist).

7. Prepare a full report:

- Create a detailed report with the system overview and list of all important files found.

8. Give expert analysis:

- Explain system structure, file details, signs of hiding or deleting data, and any important evidence discovered.

9. Provide expert support:

- Offer expert advice or courtroom testimony if required.

Types of Computer Forensics Technology

Purpose of Military Cyber Forensics

- To quickly find digital evidence.
- To know the damage caused by cyber attacks.
- To identify the attacker’s motive, identity, and location.

Challenges

- Criminals often hide, delete, or change data to avoid being caught.
- Tracking hostile activities in real time becomes difficult.

Role of NLECTC (National Law Enforcement & Corrections Technology Center)

- Works with criminal justice experts to find new technology needs.
- Tests new tools, reviews available technologies, and shares results.
- Helps bridge the gap between research and practical use.

Role of NIJ (National Institute of Justice)

- Funds research and development of new forensic tools.
- Creates best practices for cyber investigations.

Collaboration & CFX-2000

- NIJ and information directorate worked together to create CFX-2000 (Computer Forensics Experiment 2000).
- CFX-2000 is a complete forensic analysis framework.

CFX-2000 Main Idea

- Uses an integrated forensic system to identify:
- Attacker’s motive
- Purpose of attack
- Targets
- Skill level
- Identity and location

Tools Used in CFX-2000

- Uses commercial forensic software and research prototypes.
- Includes SI-FI environment (Synthesizing Information from Forensic Investigations).

What SI-FI Does

- Helps in collecting, inspecting, and analysing digital evidence.
- Stores evidence in Digital Evidence Bags (DEBs):
- Safe
- Tamper-proof
- Secure for team collaboration

Law Enforcement Computer Forensic Technology

1. Evidence Preservation

- Digital evidence is fragile and can be easily changed or deleted.
- Proper methods must be used to keep data safe during investigations.

2. Using Safe Back Software

- Safe Back creates exact mirror backups of disks.
- Helps avoid data loss and keeps evidence safe from damage or viruses.

3. Avoiding Destructive Programs

- Criminals may install programs to erase evidence or steal passwords.
- Forensic experts must detect and avoid such traps

4. Proper Documentation

- Everything must be recorded clearly.
- Important for court cases and investigations.

5. File Slack Analysis

- File slack = leftover space in a disk cluster that may still contain old evidence.
- Experts use tools to find hidden or old data here.

6. Data-Hiding Detection

- Criminals can hide files inside disks or partitions.
- Experts use tools to detect hidden data.

7. Internet & Email Activity Tracking

- Tools like Net Threat Analyzer help track past browsing and email behaviour.

8. Dual-Purpose Programs

- Some programs perform multiple actions at once (visible + hidden).
- Experts must know how these work.

9. Text Search Techniques

- Tools can search for specific words in files, slack space, or unallocated space.

10. Fuzzy Logic Tools

- Used when the exact details of stored data are unknown.
- Helps reveal hidden patterns or clues.

Data Recovery and Backup

- In today's world, information is the most valuable asset for any company. Businesses grow faster when they have quick and reliable access to their data.
- To stay competitive, companies must protect their important data.
- Special hardware and software exist for centralized data backup and recovery.
- Hardware companies make automated tape libraries that can store huge amounts of data without manual work.
- Software companies provide tools to back up and restore data from many computers using one console.
- Still, around 45% of data in networks is not backed up regularly.
- Reasons include:

- Short or tight backup windows
- Poor network design
- Lack of system administration
- Not knowing what features are required for a good backup plan
- Lack of skilled in-house staff for advanced backup systems

Without proper backups, companies risk major data loss, productivity slowdown, and huge financial damage.

Role of backup in data recovery

1. Storage Prices Are Reducing

- As disk storage becomes cheaper, more data is stored online.
- Users now expect instant access to large amounts of information.
- Restoring data from secondary storage (like tapes) is slow and less preferred.
- This impacts backup methods because people want faster recovery options.

2. System Access Must Be Online Continuously

- Many modern companies operate 24/7 (7×24).
- Huge amounts of data must always remain online and available.
- High levels of fault tolerance are needed for main data storage.

Problem: Systems cannot be taken offline for long enough to perform backups.

3. The Function of Backup Has Changed

- Backup is no longer just for recovering lost files.

- It must also:

- Protect against user errors
- Protect against data corruption
- Save clean, accurate copies that can be quickly restored
- Modern backup solutions include:
- Network-based backups
- Tape management systems
- Automated backup servers
- Tape backup systems often include:
- A workstation
- Network interface
- Tape devices
- Media server software for managing parallel backups and restore

Data recovery solutions

- Data recovery solutions ensure that an organisation can quickly restore data after failures, errors, or disasters. Modern businesses need continuous availability, so recovery planning must be strong, updated, and automated.

1. Analyse Your Preparedness

- Many organisations still use old recovery strategies created years ago.
- You must check whether your current recovery plan meets today's high-availability needs.
- Problems with outdated plans
- Old schedules like weekly backups + mid-week change logs may not be enough.
- If hundreds of logs need to be applied, recovery may take hours, which is unacceptable.
- Missing or damaged resources (logs, backups, images) can cause complete recovery failure.
- Therefore: Regularly review and update recovery strategies.

2. Prevent Resources from Slipping Through the Cracks

- A complex environment contains multiple databases, backup types, and schedules.
- Key questions:

1. Are all databases being backed up?
2. Are backups taken as frequently as planned (daily/weekly)?
3. Are change accumulations taken continuously?
4. Are there any media errors?

- If any backup resource is missing, corrupted, or delayed, recovery may fail.

3. Automated Recovery

Automation reduces the risk of human error and dependency on a single expert.

Benefits of automation:

- Less-experienced staff can perform recoveries.
- Reduces complexity in crisis scenarios.
- Helps avoid mistakes like missing logs or wrong restore points.
- Ensures faster and more accurate recovery.
- Automation makes recovery predictable, repeatable, and easier.

4. Ensure Efficient Recoveries

- Speed is crucial because downtime is expensive.

- Methods to speed up recovery:

- Multithreading: Recover multiple resources at the same time.
- Single-pass log processing: Recover multiple databases in one pass.
- Parallel tasks:
- Rebuilding indexes
- Making image copies
- Validating pointers

- These steps reduce recovery time and help the organisation resume operations quickly.

5. Take Timely and Proper Backups

- Backup is the foundation of recovery.

- Backup considerations:
- Backups should be quick, efficient, and without disturbing users.
- If downtime is available → use standard image copies.
- If systems must remain online → use advanced backup technology that works during normal operation.

- Examples:
- Online backups
- Network backups
- Incremental backups
- Automated backup servers

THANK YOU!